Calculus II — Integration

antiderivatives (indefinite integrals)

$\int f(x)\,dx = F(x)+C$ means $F'(x)=f(x)$. $+C$ accounts for all constant shifts.

Antiderivative Rules

- **Power**: $\int x^n\,dx = \frac{x^{n+1}}{n+1}+C,\ n \neq -1$
- **Constant Mult.**: $\int c f\,dx = c\int f\,dx$
- **Sum/Diff**: $\int [f \pm g]\,dx = \int f\,dx \pm \int g\,dx$
- **Reverse Chain**: $\int f(g(x))g'(x)\,dx = \int f(u)\,du, \quad u=g(x)$

Common Antiderivatives

- $\int 0\,dx = C, \quad \int 1\,dx = x+C$
- $\int x^n\,dx = \frac{x^{n+1}}{n+1}+C \quad (n \neq -1)$
- **ln-flag**: $\int \frac{1}{x}\,dx = \int x^{-1}\,dx = \ln|x|+C$. Any time you see u^{-1} (power $=-1$), it's a ln, *never* the power rule (that divides by 0).
- $\int e^x\,dx = e^x+C, \quad \int a^x\,dx = \frac{a^x}{\ln(a)}+C$
- $\int \sin(x)\,dx = -\cos(x)+C$
- $\int \cos(x)\,dx = \sin(x)+C$
- $\int \sec^2(x)\,dx = \tan(x)+C$
- $\int \csc^2(x)\,dx = -\cot(x)+C$
- $\int \sec(x)\tan(x)\,dx = \sec(x)+C$
- $\int \csc(x)\cot(x)\,dx = -\csc(x)+C$
- $\int \sec(x)\,dx = \ln|\sec x+\tan x|+C$
- $\int \frac{1}{\sqrt{1-x^2}}\,dx = \arcsin(x)+C$
- $\int \frac{1}{1+x^2}\,dx = \arctan(x)+C$

Inside a constant (k) — the $\frac{1}{k}$ rule

When the inside is kx instead of x, divide by the inside's derivative k (mini u-sub):

- $\int e^{kx}\,dx = \frac{1}{k}e^{kx}+C$
- $\int \cos(kx)\,dx = \frac{1}{k}\sin(kx)+C$
- $\int \sin(kx)\,dx = -\frac{1}{k}\cos(kx)+C$

Any *linear* inside $kx+b$ works the same — the shift changes nothing:
$\int e^{3x-1}\,dx = \frac{1}{3}e^{3x-1}+C$; $\int (2x+5)^7\,dx = \frac{(2x+5)^8}{16}+C$;
$\int \frac{dx}{4x+3} = \frac{1}{4}\ln|4x+3|+C$ *(the ln-flag wearing its $\frac{1}{k}$).*

Arctan / arcsin spotting

A lone constant $+x^2$ in a denominator (no x in the numerator) is an arctan; a constant $-x^2$ under a root is an arcsin. General forms:

$$\int \frac{1}{a^2+x^2}\,dx = \frac{1}{a}\arctan\left(\frac{x}{a}\right)+C \qquad \int \frac{1}{\sqrt{a^2-x^2}}\,dx = \arcsin\left(\frac{x}{a}\right)+C$$

Worked: arctan — $\int \frac{1}{4+x^2}\,dx$

1. MATCH THE FORM
constant $+x^2$, no x upstairs: arctan. $a^2=4 \Rightarrow a=2$

↓

2. APPLY — WITH THE $\frac{1}{a}$
$\frac{1}{2}\arctan\left(\frac{x}{2}\right)+C$ *(the $\frac{1}{a}$ out front is the most common miss; $a=1$ gives the plain $\arctan x$)*

Worked: arcsin twin — $\int \frac{1}{\sqrt{9-x^2}}\,dx$

1. MATCH THE FORM
constant $-x^2$ under a root: arcsin. $a^2=9 \Rightarrow a=3$

↓

2. APPLY — NO $\frac{1}{a}$ THIS TIME
$\arcsin\left(\frac{x}{3}\right)+C$ *(arctan keeps a $\frac{1}{a}$, arcsin doesn't — the root supplies it)*

riemann sums

Partition $[a,b]$ into n equal subintervals of width $\Delta x = \frac{b-a}{n}$. Pick x_i^* in each:
Area $\approx \sum_{i=1}^{n} f(x_i^*)\cdot\Delta x$

Sample Point Choices

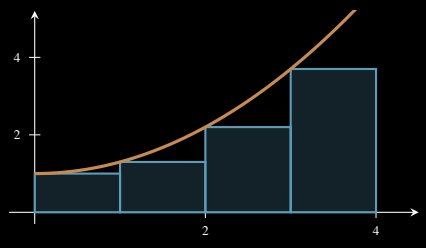
- **Left** L_n: $x_i = a+i\Delta x$, $i=0,\ldots,n-1$
- **Right** R_n: $x_i = a+i\Delta x$, $i=1,\ldots,n$
- **Midpoint** M_n: $x_i = a+(i-\frac{1}{2})\Delta x$
- **Trapezoidal**: $T_n = \frac{L_n+R_n}{2}$

Summation Formulas

- $\sum_{i=1}^{n} c = cn, \quad \sum [ca_i] = c\sum a_i$
- $\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$
- $\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}$
- $\sum_{i=1}^{n} i^3 = \left[\frac{n(n+1)}{2}\right]^2$
- $\sum [a_i \pm b_i] = \sum a_i \pm \sum b_i$

Example: $\sum_{i=1}^{10}(i^2+3i) = \sum i^2+3\sum i = 385+3(55) = 550$ — split, pull constants, plug the closed forms.

L_4 for $f(x)=0.3x^2+1$ on $[0,4]$



Worked: the drawn L_4, computed — then R_4, M_4

1. WIDTH & SAMPLE HEIGHTS
$f(x)=0.3x^2+1$, $\Delta x = \frac{4-0}{4}=1$; left heights: $f(0)=1$, $f(1)=1.3$, $f(2)=2.2$, $f(3)=3.7$

↓

2. SUM × WIDTH
$L_4 = (1+1.3+2.2+3.7)(1) = 8.2$

↓

3. RIGHT SAMPLES — DROP $f(0)$, ADD $f(4)$
$f(4)=0.3(16)+1=5.8$: $R_4 = (1.3+2.2+3.7+5.8)(1) = 13$

↓

4. MIDPOINT SAMPLES $x=0.5, 1.5, 2.5, 3.5$
$f = 1.075, \quad 1.675, \quad 2.875, \quad 4.675$ *(for example, $f(0.5)=0.3(0.25)+1$)*:
$M_4 = (1.075+1.675+2.875+4.675)(1) = 10.3$

↓

5. OVER OR UNDER?
exact: $\int_0^4 (0.3x^2+1)dx = \left[0.1x^3+x\right]_0^4 = (6.4+4)-0 = 10.4$: f increasing $\Rightarrow$ L under, R over — always; M lands closest

sigma to integral

$\int_a^b f(x)\,dx = lim_{n\to\infty}\sum_{i=1}^{n} f(x_i^*)\Delta x$
$\Sigma \to \int$, $\Delta x \to dx$, $x_i^* \to x$. Bounds carry over.

Worked: evaluate $\int_0^2 x^2\,dx$ from the definition

1. BUILD THE RIGHT SUM
$\Delta x = \frac{2}{n}$, $x_i = \frac{2i}{n}$: $\sum_{i=1}^{n}\left(\frac{2i}{n}\right)^2\frac{2}{n} = \sum_{i=1}^{n}\frac{4i^2}{n^2}\cdot\frac{2}{n} = \frac{8}{n^3}\sum_{i=1}^{n} i^2$ *(constants slide out of the Σ)*

↓

2. CLOSED FORM
$\frac{8}{n^3}\cdot\frac{n(n+1)(2n+1)}{6} = \frac{8(n+1)(2n+1)}{6n^2}$

↓

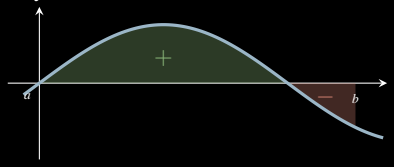
3. LIMIT
$n\to\infty$: leading terms $\dfrac{8\cdot 2n^2}{6n^2}=\dfrac{8}{3}$ *(FTC agrees: $\frac{x^3}{3}\big|_0^2=\frac{8}{3}$)*

Worked: reverse — recognize the integral

$\lim\limits_{n\to\infty}\sum\limits_{i=1}^{n}\left(1+\frac{2i}{n}\right)^3\frac{2}{n}$: read off $\Delta x=\frac{2}{n}\Rightarrow b-a=2$; $x_i=1+\frac{2i}{n}\Rightarrow a=1$; the function wraps x_i cubed $\Rightarrow \int_1^3 x^3\,dx$.

definite integral

Net signed area between $y=f(x)$ and x-axis on $[a,b]$. Below x-axis contributes negatively.



Properties

- $\int_a^a f\,dx=0$, $\int_b^a f\,dx=-\int_a^b f\,dx$
- $\int_a^b c\,dx=c(b-a)$
- $\int_a^b cf\,dx=c\int_a^b f\,dx$
- $\int_a^b[f\pm g]\,dx=\int_a^b f\pm\int_a^b g$
- $\int_a^b f\,dx=\int_a^c f\,dx+\int_c^b f\,dx$
- $f\ge 0\Rightarrow\int_a^b f\,dx\ge 0$
- $f\ge g\Rightarrow\int_a^b f\ge\int_a^b g$

Example: given $\int_0^5 f=10$ and $\int_0^3 f=4$: $\int_3^5 f=10-4=6$ (additivity), and $\int_5^3 f=-6$ (reversal).

fundamental theorem of calculus

FTC Part 1 (derivative of an integral)

If f continuous on $[a,b]$, $F(x)=\int_a^x f(t)\,dt$ is differentiable:
$F'(x)=\frac{d}{dx}\int_a^x f(t)\,dt=f(x)$
With chain (variable upper bound):
$\frac{d}{dx}\int_a^{g(x)} f(t)\,dt=f(g(x))\cdot g'(x)$
Both bounds variable:
$\frac{d}{dx}\int_{h(x)}^{g(x)} f(t)\,dt=f(g(x))g'(x)-f(h(x))h'(x)$

Worked: all three variants on $f(t)=\cos(t^2)$

1. PLAIN
$\frac{d}{dx}\int_1^x\cos(t^2)\,dt=\cos(x^2)$ *(stamp x in for t — no integration happens)*

↓

2. CHAIN BOUND
$\frac{d}{dx}\int_0^{x^3}\cos(t^2)\,dt=\cos\big((x^3)^2\big)\cdot 3x^2=3x^2\cos(x^6)$

↓

3. VARIABLE *LOWER* BOUND — SIGN FLIP
$\frac{d}{dx}\int_x^5\cos(t^2)\,dt=\frac{d}{dx}\left[-\int_5^x\right]=-\cos(x^2)$ *(the most-missed variant)*

FTC Part 2 (evaluation)

If f continuous and F any antiderivative of f:
$\int_a^b f(x)\,dx=F(b)-F(a)=F(x)\big|_a^b$
Example: $\int_1^4 3\sqrt{x}\,dx=3\cdot\frac{2}{3}x^{3/2}\big|_1^4=2(8-1)=14$.

Absolute value integrals

Split at roots of the inside, then determine sign on each piece.

$$|f(t)|=\begin{cases} f(t) & f(t)\ge 0\\ -f(t) & f(t)<0\end{cases}$$

Worked: $\int_{-2}^{4}|t^2-2t-3|\,dt$

1. ROOTS OF THE INSIDE
$(t-3)(t+1)=0\Rightarrow t=-1,\,3$ — both inside $[-2,4]$

↓

2. SPLIT & TEST A POINT PER PIECE
$t=-1.5\to+$, $t=0\to-$, $t=3.5\to+$ $\Rightarrow$ signs $+-+$

↓

3. FLIP THE NEGATIVE PIECE
$\int_{-2}^{-1}f-\int_{-1}^{3}f+\int_{3}^{4}f$, $f=t^2-2t-3$

↓

4. EVALUATE WITH $F=\frac{t^3}{3}-t^2-3t$
$[F]_{-2}^{-1}=\frac{5}{3}-(-\frac{2}{3})=\frac{7}{3}$; $-[F]_{-1}^{3}=-(-9-\frac{5}{3})=\frac{32}{3}$; $[F]_3^4=-\frac{20}{3}-(-9)=\frac{7}{3}$

↓

5. ADD
$\frac{7}{3}+\frac{32}{3}+\frac{7}{3}=\frac{46}{3}$

Tip: Upward parabolas give $+-+$ across both roots. Downward give $-+-$.

u-substitution

Reverses the chain rule. Pick the "inside" function as u; its derivative should appear (up to a constant) elsewhere in the integrand.

Indefinite integral steps

1. **Pick** $u=g(x)$ (usually the inside of a composition, or whatever has its derivative present).
2. **Differentiate**: $\frac{du}{dx}=g'(x)$, so $du=g'(x)\,dx$.
3. **Solve for** dx: $dx=\frac{du}{g'(x)}$ (or rearrange to substitute directly).
4. **Substitute**: replace every x-piece with a u-piece. The integral should be entirely in u.
5. **Integrate** in u.
6. **Back-substitute**: replace u with $g(x)$. Add $+C$.

General form

$\int f(g(x))\cdot g'(x)\,dx=\int f(u)\,du=F(u)+C=F(g(x))+C$

Definite integral: change the bounds

1. Pick $u=g(x)$, find du.
2. **Convert bounds**: new lower $=g(a)$, new upper $=g(b)$.
3. Integrate in u and evaluate at new bounds — *no back-substitution needed.*

$\int_a^b f(g(x))\cdot g'(x)\,dx=\int_{g(a)}^{g(b)} f(u)\,du$

Color key: $\boxed{u}$ = the u-piece (cyan), $\boxed{du}$ = the du-piece (orange). Find both in the original, then swap them across.

Example (indefinite): $\int 2x\cos(x^2)\,dx$

Spot the $\boxed{\text{inside}}$ and its $\boxed{\text{derivative}}$, then walk each step down its own box:

$$\int \boxed{2x}\cos(\boxed{x^2})\,\boxed{dx}$$

1. PICK u
$\boxed{u}=\boxed{x^2}$ *(the inside of* cos*)*

↓

2. DIFFERENTIATE
$\boxed{du}=\boxed{2x\,dx}$ *(already sitting in the integrand)*

↓

3. SUBSTITUTE
$\boxed{x^2}\to\boxed{u}$, $\boxed{2x\,dx}\to\boxed{du}$: $\int\cos(\boxed{u})\,\boxed{du}$

↓

4. INTEGRATE in u
$\int\cos(\boxed{u})\,\boxed{du}=\sin(\boxed{u})+C$

↓

5. BACK-SUB
$\boxed{u} \to \boxed{x^2}$: $\sin(x^2)+C$

Example (definite): $\int_0^2 x e^{x^2}\,dx$

Same swap, but *convert the bounds* instead of back-substituting:

$$\int_0^2 \boxed{x}\, e^{\boxed{x^2}}\, \boxed{dx}$$

1. PICK u
$\boxed{u}=\boxed{x^2}$

↓

2. DIFFERENTIATE
$\boxed{du}=\boxed{2x\,dx} \Rightarrow \boxed{x\,dx}=\frac{1}{2}\boxed{du}$ *(off by a constant)*

↓

3. CONVERT BOUNDS
$x=0 \to \boxed{u}=0, \quad x=2 \to \boxed{u}=4$

↓

4. INTEGRATE & EVALUATE
$\frac{1}{2}\int_0^4 e^{\boxed{u}}\,\boxed{du} = \frac{1}{2}e^{\boxed{u}}\Big|_0^4 = \frac{1}{2}(e^4-1)$

No back-substitution once bounds are converted — evaluate straight in u at the new bounds.

Worked (ln flavor): $\int \frac{2x}{x^2+1}\,dx$

1. SPOT THE PAIR
the denominator's derivative sits in the numerator: $\boxed{u}=\boxed{x^2+1}, \quad \boxed{du}=\boxed{2x\,dx}$

↓

2. SUBSTITUTE
$\int \frac{\boxed{du}}{\boxed{u}}$ — the power -1 case: ln, never the power rule

↓

3. INTEGRATE & BACK-SUB
$\ln|u|+C = \ln(x^2+1)+C$

Worked (leftover x): $\int x\sqrt{x-1}\,dx$

1. PICK u
$\boxed{u}=\boxed{x-1}, \ \boxed{du}=\boxed{dx}$ — but a naked x remains

↓

2. REWRITE THE LEFTOVER IN u
$x=u+1$: $\int (u+1)\sqrt{u}\,du = \int (u^{3/2}+u^{1/2})\,du$

↓

3. INTEGRATE & BACK-SUB
$\frac{2}{5}u^{5/2}+\frac{2}{3}u^{3/2}+C = \frac{2}{5}(x-1)^{5/2}+\frac{2}{3}(x-1)^{3/2}+C$

Picking u — quick checklist

- Inside of a composition (under a root, in an exponent, in a trig function).
- Inside of parentheses raised to a power.
- Denominator (when its derivative is in the numerator).
- Whichever piece's derivative is "lying around" (up to a constant multiple — pull it out via step 3).

Constant multiple trick

If du is off by a constant, solve for what you need:
$du=6x\,dx \Rightarrow x\,dx=\frac{1}{6}du$
Pull the $\frac{1}{6}$ out front of the integral.

When u-sub fails

Try integration by parts, trig identities to simplify first, partial fractions, or trig substitution — all worked below. If u doesn't eliminate *all* x's, the choice was wrong.

integration by parts (IBP)

For a *product* of two unrelated functions where no inside-derivative is lying around (so u-sub won't work):

$$\int u\,dv = uv - \int v\,du$$

Pick u by LIATE

Choose u as the type that comes *first* in this list (it gets simpler when differentiated); the rest is dv.

L	Logarithm	$\ln x$
I	Inverse trig	$\arctan x$, $\arcsin x$
A	Algebraic	x, x^2
T	Trig	$\sin x$, $\cos x$
E	Exponential	e^x

4-step process

1. **Pick** u (LIATE) and dv (everything else, including dx).
2. **Build** du (differentiate u) and v (antiderivative of dv).
3. **Plug in**: $uv - \int v\,du$.
4. **Solve the NEW integral** $\int v\,du$ — it's a *separate* antiderivative, not "done" just by substituting.

IBP trades a hard integral for an easier one. If the new integral is worse, swap your u/dv choice.

Worked: $\int x e^x\,dx$

1. PICK BY LIATE, BUILD THE PARTNERS

PICK	BUILD
$\boxed{u}=\boxed{x}$	$\boxed{du}=\boxed{dx}$ *(A before E $\Rightarrow u=x$)*
$dv=e^x\,dx$	$v=e^x$

↓

2. PLUG INTO $uv-\int v\,du$
$\boxed{x}\,e^x - \int e^x\,\boxed{dx}$

↓

3. SOLVE THE NEW INTEGRAL
$xe^x - e^x = e^x(x-1)+C$ *(differentiating x down to 1 is what made it easy)*

Worked: the L case — $\int \ln x\,dx$

1. THE $dv=dx$ TRICK
no visible product? manufacture one: $u=\ln x, \quad dv=dx \Rightarrow du=\frac{1}{x}dx, \quad v=x$

↓

2. PLUG IN
$x\ln x - \int x\cdot\frac{1}{x}\,dx = x\ln x - \int 1\,dx$

↓

3. FINISH
$x\ln x - x + C$

The I case is identical: $\int \arctan x\,dx$ with $u=\arctan x$, $dv=dx$: $x\arctan x - \int \frac{x}{1+x^2}dx$; the new integral is the ln-flavor u-sub, off by a constant: $\int \frac{x}{1+x^2}dx = \frac{1}{2}\int \frac{2x}{1+x^2}dx = \frac{1}{2}\ln(1+x^2) \ \Rightarrow \ x\arctan x - \frac{1}{2}\ln(1+x^2)+C$.

Worked: repeated IBP — $\int x^2 e^x\,dx$

1. IBP #1 — BUILD, PLUG, PULL THE 2
$u=x^2, du=2x\,dx; \quad dv=e^x dx, v=e^x$: $x^2e^x - \int e^x\cdot 2x\,dx = x^2e^x - 2\int xe^x\,dx$

↓

2. THE NEW INTEGRAL IS THE FIRST WORKED
$\int xe^x\,dx = e^x(x-1)$

↓

3. ASSEMBLE
$x^2e^x - 2e^x(x-1) = e^x(x^2-2x+2)+C$ *(each A-round drops the power by one — repeat until it dies)*

Loop trick ($e^x\sin x$, $\sin(\ln x)$, $e^x\cos x\ldots$)

When IBP twice brings the *original* integral back on the right:
1. Apply IBP twice (stop — don't do a third).
2. The original integral I reappears: you get $I=(\text{stuff})-I$.
3. Collect: $2I=\text{stuff}$, then divide by 2.

Worked: loop — $I=\int e^x\sin x\,dx$

1. IBP #1
$u=\sin x$, $dv=e^x dx$: $\quad I=e^x\sin x-\int e^x\cos x\,dx$

↓

2. IBP #2 — SAME PATTERN, WATCH THE SIGN
$u=\cos x$, $du=-\sin x\,dx$, $v=e^x$:

$$\int e^x\cos x\,dx=e^x\cos x-\int e^x(-\sin x)dx=e^x\cos x+\int e^x\sin x\,dx$$

$$\Rightarrow\ I=e^x\sin x-\left[e^x\cos x+\int e^x\sin x\,dx\right]$$

(stop — the original I just walked back in)

↓

3. COLLECT THE LOOP
$I=e^x\sin x-e^x\cos x-I\ \Rightarrow\ 2I=e^x(\sin x-\cos x)$

↓

4. DIVIDE BY 2
$I=\dfrac{e^x(\sin x-\cos x)}{2}+C$

The same loop (with $dv=dx$ both rounds) closes $I=\int\sin(\ln 2x)\,dx$: round 1 gives $I=x\sin(\ln 2x)-\int\cos(\ln 2x)\,dx$; round 2 turns the new integral into $x\cos(\ln 2x)+I$, so $I=x\sin(\ln 2x)-x\cos(\ln 2x)-I\ \Rightarrow$ $2I=x[\sin(\ln 2x)-\cos(\ln 2x)]\ \Rightarrow\ I=\frac{x[\sin(\ln 2x)-\cos(\ln 2x)]}{2}+C$.

Helper antiderivatives: $\int\cos(kx)dx=\frac{1}{k}\sin(kx)$, $\quad\int\sin(kx)dx=-\frac{1}{k}\cos(kx)$.

IBP vs u-sub — which one?

- u-sub: a composition *and* its derivative is present elsewhere.
- IBP: a product of unrelated functions, no inside-derivative.

trig integrals

Identities you'll need:
- $\sin^2x+\cos^2x=1$, $\quad 1+\tan^2x=\sec^2x$, $\quad 1+\cot^2x=\csc^2x$
- $\sin 2x=2\sin x\cos x$, $\quad\cos 2x=\cos^2x-\sin^2x$
- $\sin^2x=\frac{1-\cos 2x}{2}$, $\quad\cos^2x=\frac{1+\cos 2x}{2}$

***Odd power of* sin *or* cos**

Peel *one* factor off the odd power, convert the rest with $\cos^2=1-\sin^2$ (or $\sin^2=1-\cos^2$), then u-sub.

Worked (odd cos): $\int\sqrt{\sin x}\cos^3x\,dx$

1. PEEL ONE OFF THE ODD POWER
$\cos^3x=(1-\sin^2x)\cos x$

↓

2. U-SUB
$\boxed{u}=\boxed{\sin x}$, $\quad\boxed{du}=\boxed{\cos x\,dx}$ — the peeled factor *is* the du

↓

3. INTEGRATE IN u
$\int u^{1/2}(1-u^2)\,du=\int u^{1/2}-u^{5/2}\,du=\frac{2}{3}u^{3/2}-\frac{2}{7}u^{7/2}$

↓

4. BACK-SUB
$\frac{2}{3}(\sin x)^{3/2}-\frac{2}{7}(\sin x)^{7/2}+C$

Worked (odd sin — the minus-sign case): $\int\sin^3x\cos^2x\,dx$

1. PEEL
$\sin^3x=(1-\cos^2x)\sin x$

↓

2. U-SUB — WATCH THE SIGN
$u=\cos x$, $\quad du=-\sin x\,dx$ *(the classic error lives here)*

↓

3. INTEGRATE
$-\int(1-u^2)u^2\,du=-\int(u^2-u^4)\,du=-\left(\frac{u^3}{3}-\frac{u^5}{5}\right)=-\frac{u^3}{3}+\frac{u^5}{5}$

↓

4. BACK-SUB
$\frac{\cos^5x}{5}-\frac{\cos^3x}{3}+C$

Even powers — power reduction

No factor to peel, so drop the power with:

$$\sin^2x=\frac{1-\cos 2x}{2}\qquad\cos^2x=\frac{1+\cos 2x}{2}$$

Worked: $\int\sin^2x\,dx$, then $\int\sin^4x\,dx$

1. REDUCE
$\int\frac{1-\cos 2x}{2}dx=\frac{1}{2}\int 1\,dx-\frac{1}{2}\int\cos 2x\,dx=\frac{x}{2}-\frac{1}{2}\cdot\frac{\sin 2x}{2}=\frac{x}{2}-\frac{\sin 2x}{4}+C$

↓

2. FOURTH POWER — SQUARE THE REDUCTION
$\sin^4x=\left(\frac{1-\cos 2x}{2}\right)^2=\frac{1-2\cos 2x+\cos^2 2x}{4}$ *(FOIL first — $(1-\cos 2x)^2$ can't be integrated whole)*

↓

3. REDUCE THE LEFTOVER SQUARE
$\cos^2 2x=\frac{1+\cos 4x}{2}$:

$$\sin^4x=\frac{1-2\cos 2x+\frac{1+\cos 4x}{2}}{4}=\frac{\frac{3}{2}-2\cos 2x+\frac{\cos 4x}{2}}{4}=\frac{3}{8}-\frac{\cos 2x}{2}+\frac{\cos 4x}{8}$$

↓

4. INTEGRATE TERMWISE
$\frac{3x}{8}-\frac{\sin 2x}{4}+\frac{\sin 4x}{32}+C$

sec/tan family

$\int\tan x\,dx=\int\frac{\sin x}{\cos x}dx$ — ln-flavor u-sub ($u=\cos x$): $-\ln|\cos x|+C=\ln|\sec x|+C$.
- **Even sec power**: peel $\sec^2x$, convert the rest via $\sec^2=1+\tan^2$, $\quad u=\tan x$.
- **Odd tan power (with sec)**: peel $\sec x\tan x$, convert via $\tan^2=\sec^2-1$, $u=\sec x$.
- $\int\sec^3x\,dx$: IBP loop trick $\Rightarrow\ \frac{1}{2}\left[\sec x\tan x+\ln|\sec x+\tan x|\right]+C$.

Product-to-sum: $\int\sin(3x)\cos(x)\,dx$

1. CONVERT THE PRODUCT
$\sin A\cos B=\frac{1}{2}[\sin(A+B)+\sin(A-B)]$: $\quad\frac{1}{2}[\sin 4x+\sin 2x]$

↓

2. INTEGRATE TERMWISE
$-\frac{\cos 4x}{8}-\frac{\cos 2x}{4}+C$

partial fractions

For rational integrands where u-sub fails: split into small pieces that are each a ln or a power. Requires $deg\,(\text{top})<deg\,(\text{bottom})$ — long-divide first if not.
Setup by denominator factor type:
- **Distinct linear** $(x-a)(x-b)$: $\quad\frac{A}{x-a}+\frac{B}{x-b}$
- **Repeated** $(x-a)^2$: $\quad\frac{A}{x-a}+\frac{B}{(x-a)^2}$ — one term per power
- **Irreducible quadratic**: $\quad\frac{Ax+B}{x^2+px+q}$

Worked: $\int\frac{x+7}{(x-1)(x+3)}dx$

1. BUILD THE SPLIT
$\frac{x+7}{(x-1)(x+3)}=\frac{A}{x-1}+\frac{B}{x+3}\ \Rightarrow\ x+7=A(x+3)+B(x-1)$

↓

2. KILL EACH UNKNOWN WITH A ROOT
$x=1$: $8=4A\Rightarrow A=2$. $\quad x=-3$: $4=-4B\Rightarrow B=-1$ *(plug each root — the other term vanishes)*

↓

3. INTEGRATE THE PIECES
$\int\frac{2}{x-1}-\frac{1}{x+3}dx=2\ln|x-1|-\ln|x+3|+C$

Worked (repeated factor): $\int \frac{3x+5}{(x+1)^2}\,dx$

1. ONE TERM PER POWER
$\frac{3x+5}{(x+1)^2}=\frac{A}{x+1}+\frac{B}{(x+1)^2} \;\Rightarrow\; 3x+5=A(x+1)+B$

↓

2. SOLVE
$x=-1$: $B=2$; match x-coefficients: $A=3$

↓

3. INTEGRATE
$\int \frac{3}{x+1}+\frac{2}{(x+1)^2}\,dx=3\ln|x+1|-\frac{2}{x+1}+C$ *(second piece is the power rule, not a ln)*

trig substitution

For roots of quadratics that u-sub can't crack — substitute a trig function so the Pythagorean identity collapses the root:

sees	substitute	root becomes
$\sqrt{a^2-x^2}$	$x=a\sin\theta$	$a\cos\theta$
$\sqrt{a^2+x^2}$	$x=a\tan\theta$	$a\sec\theta$
$\sqrt{x^2-a^2}$	$x=a\sec\theta$	$a\tan\theta$

Worked: $\int \frac{dx}{x^2\sqrt{x^2+4}}$

1. PICK FROM THE TABLE
a^2+x^2 shape, $a=2$: $x=2\tan\theta$, $dx=2\sec^2\theta\,d\theta$, $\sqrt{x^2+4}=2\sec\theta$

↓

2. SUBSTITUTE EVERYTHING
$\int \frac{2\sec^2\theta\,d\theta}{4\tan^2\theta\cdot 2\sec\theta}=\frac{1}{4}\int\frac{\sec\theta}{\tan^2\theta}\,d\theta=\frac{1}{4}\int\frac{\cos\theta}{\sin^2\theta}\,d\theta$

↓

3. EASY TRIG INTEGRAL
$u=\sin\theta$, $du=\cos\theta\,d\theta$: $\frac{1}{4}\int\frac{du}{u^2}=\frac{1}{4}\int u^{-2}du=\frac{1}{4}\left(-\frac{1}{u}\right)=-\frac{1}{4\sin\theta}$

↓

4. TRIANGLE BACK-CONVERT
$\tan\theta=\frac{x}{2}\Rightarrow$ draw the right triangle: opp x, adj 2, hyp $\sqrt{x^2+4}$ $\Rightarrow$ $\sin\theta=\frac{x}{\sqrt{x^2+4}}$:
$-\frac{\sqrt{x^2+4}}{4x}+C$

net change theorem

The integral of a rate of change is the total change:

$$\int_a^b f'(x)\,dx=f(b)-f(a)$$

Position from velocity: integrate $v(t)$ to get $s(t)+C$, then use a given condition (for example, $s(0)=s_0$) to solve for C.

Worked: displacement vs total distance — $v=(t-1)(t-3)$ ***on*** $[0,4]$

1. DISPLACEMENT — JUST INTEGRATE
$\int_0^4 (t^2-4t+3)\,dt=\left[\frac{t^3}{3}-2t^2+3t\right]_0^4=\left(\frac{64}{3}-32+12\right)-0=\frac{64}{3}-20=\frac{4}{3}$

↓

2. DISTANCE — INTEGRATE $|v|$
$v<0$ on $(1,3)$: split there. With $F=\frac{t^3}{3}-2t^2+3t$: $F(0)=0$, $F(1)=\frac{4}{3}$, $F(3)=9-18+9=0$, $F(4)=\frac{4}{3}$

↓

3. PIECES — $F(b)-F(a)$ EACH
$\int_0^1 v=\frac{4}{3}-0=\frac{4}{3}$; $\int_1^3 v=0-\frac{4}{3}=-\frac{4}{3}\xrightarrow{|\cdot|}\frac{4}{3}$; $\int_3^4 v=\frac{4}{3}-0=\frac{4}{3}$ $\Rightarrow$ distance $=4$

Backing up subtracts from displacement ($\frac{4}{3}$) but adds to distance (4) — two different questions, one $|\cdot|$ apart.

average value

$f_{avg}=\frac{1}{b-a}\int_a^b f(x)\,dx$
MVT for Integrals: f cont. on $[a,b]\Rightarrow\exists c\in[a,b]$ with $f(c)=f_{avg}$.

Worked: $f=x^2$ ***on*** $[0,3]$ ***— average, then the*** c ***that attains it***

1. AVERAGE VALUE
$\int_0^3 x^2\,dx=\left[\frac{x^3}{3}\right]_0^3=\frac{27}{3}-0=9$: $f_{avg}=\frac{1}{3-0}\cdot 9=3$

↓

2. MVT-FOR-INTEGRALS FOLLOW-UP
solve $f(c)=f_{avg}$: $c^2=3\Rightarrow c=\sqrt{3}$ *(keep only roots inside $[0,3]$ — discard $-\sqrt{3}$)*

area between curves

$$A=\int_a^b[\underbrace{\text{top}}_{f}-\underbrace{\text{bottom}}_{g}]\,dx \qquad A=\int_c^d[\underbrace{\text{right}}-\underbrace{\text{left}}]\,dy$$

Left form uses vertical slices (dx); right form uses horizontal slices (dy).

THE PROCESS
1. Sketch both curves.
2. Find every intersection. Solve $f(x)=g(x)$.
3. Choose the slice direction — vertical (dx) or horizontal (dy).
4. Integrate the gap: (top−bottom)$\,dx$, or (right−left)$\,dy$.

Step 3 decoded — three region shapes

A dx — one curve stays on top across the whole interval. $\int_a^b$ (top−bottom)$\,dx$
B dy — the region reads better left-to-right. Solve both curves for x first, then $\int_c^d$ (right−left)$\,dy$
C **curves cross inside** the interval $\Rightarrow$ split at each crossing. The top curve changes there.

THE THREE THINGS YOU SOLVE
1. Intersections — set $f(x)=g(x)$ and solve. These are the limits, or the split points.
2. Which curve is on top — test one interior point. Do not guess from the formulas.
3. Solve for x when you slice horizontally: $y=x^2\Rightarrow x=\sqrt{y}$.

4-step process

1. **Bounds**: set curves equal, solve for intersections (FOIL-check your factoring).
2. **Test a point**: which curve is top/bottom (or right/left)?
3. **Set up**: bigger − smaller.
4. **Evaluate** (or stop if "setup only").

Vertical (dx) vs horizontal (dy)?

Ask: *"Does any boundary switch partway through the region?"*
- Bottom switches at some $x\Rightarrow dx$ needs 2 integrals $\Rightarrow$ switch to dy.
- A smooth curve like $y=x^2$ does *not* force a switch. Two lines meeting at a corner does.

Integrating along y ***— solve for*** x

Bounds are y-values, integrand is right−left, both as functions of y, end with dy.
- $y=x^2\Rightarrow x=\pm\sqrt{y}$ (take the side in your region)
- $y=\sqrt{x}\Rightarrow x=y^2$, $y=x+2\Rightarrow x=y-2$

Worked: $y=x+2$ ***and*** $y=x^2$

1. BOUNDS — SET EQUAL
$x+2=x^2\Rightarrow x^2-x-2=(x-2)(x+1)=0\Rightarrow x=-1,2$

↓

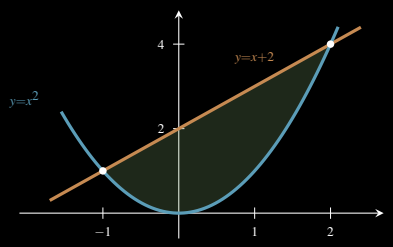
2. TEST A POINT
$x=0$: line 2 vs parabola 0 — line on top

↓

3. BUILD THE INTEGRAL, TOP − BOTTOM
$A=\int_{-1}^{2}\left[(x+2)-x^2\right]dx$ *(the minus hits every term)*

↓

4. EVALUATE
$\left[\frac{x^2}{2}+2x-\frac{x^3}{3}\right]_{-1}^{2}=\frac{10}{3}-\left(-\frac{7}{6}\right)=\frac{9}{2}$



Worked (dy): $x=y^2$ and $x=y+2$

The mirror-image region — and here dx would need two integrals (the bottom boundary switches from parabola to line at $x=1$), while dy needs one:

1. INTERSECT IN y
$y^2=y+2 \Rightarrow (y-2)(y+1)=0 \Rightarrow y=-1, 2$

↓

2. RIGHT MINUS LEFT
$y=0$: line $x=2$ vs parabola $x=0$ — line is right: $(y+2)-y^2$

↓

3. BUILD THE INTEGRAL, THEN INTEGRATE IN y
$A=\int_{-1}^{2}\left[(y+2)-y^2\right]dy=\left[\frac{y^2}{2}+2y-\frac{y^3}{3}\right]_{-1}^{2}=\frac{10}{3}-\left(-\frac{7}{6}\right)=\frac{9}{2}$ *(same $\frac{9}{2}$ as above — the same problem with $x \leftrightarrow y$)*

Worked (curves that cross): $y=x$ and $y=x^3$ on $[-1,1]$

1. FIND ALL CROSSINGS
$x^3=x \Rightarrow x(x^2-1)=0 \Rightarrow x=-1, 0, 1$ — a crossing inside means one integral is wrong

↓

2. TOP PER PIECE
$x=-\frac{1}{2}$: $x^3=-\frac{1}{8}>-\frac{1}{2}$ — cubic on top; $x=\frac{1}{2}$: line on top

↓

3. ONE INTEGRAL PER PIECE
$\int_{-1}^{0}(x^3-x)dx=\left[\frac{x^4}{4}-\frac{x^2}{2}\right]_{-1}^{0}=0-\left(\frac{1}{4}-\frac{1}{2}\right)=\frac{1}{4}$

$\int_{0}^{1}(x-x^3)dx=\left[\frac{x^2}{2}-\frac{x^4}{4}\right]_{0}^{1}=\frac{1}{2}-\frac{1}{4}=\frac{1}{4}$; total $\frac{1}{2}$

geometry as integrals

- Rectangle ($f=c$): $A=c(b-a)$
- Triangle (linear f): $A=\frac{1}{2}bh$
- Semicircle $\sqrt{r^2-(x-h)^2}$: $A=\frac{1}{2}\pi r^2$
- Quarter circle: $A=\frac{1}{4}\pi r^2$

Radius from $\sqrt{A-Bx^2}$: factor B to get $\sqrt{B}\sqrt{A/B-x^2}$, so $r=\sqrt{A/B}$.

volumes of revolution

Spin a region around an axis → solid. Pick the method by what the slice looks like.

Method	Slice	Volume
Disc	solid circle	$\pi\int_a^b [R]^2\,dx$
Washer	circle w/ hole	$\pi\int_a^b ([R]^2-[r]^2)\,dx$
Shell	cylinder peel	$2\pi\int_a^b (\text{rad})(\text{ht})\,dx$

Disc vs washer: region touches the axis ⇒ disc ($r=0$); gap between region and axis ⇒ washer. Square *each* radius separately, *then* subtract.

THE PROCESS
1. **Sketch** the region and draw the axis.
2. **Choose the slice** — perpendicular to the axis (disc or washer), or parallel to it (shell).
3. **Write each radius** as a distance from the axis.
4. **Integrate.**

Step 2 decoded — disc, washer or shell?

A **disc** — slice ⊥ axis, and the region touches the axis: $V=\pi\int R^2$

B **washer** — slice ⊥ axis, but a gap sits between region and axis: $V=\pi\int (R_{\text{out}}^2-R_{\text{in}}^2)$

C **shell** — slice ∥ axis: $V=2\pi\int (\text{radius})(\text{height})$

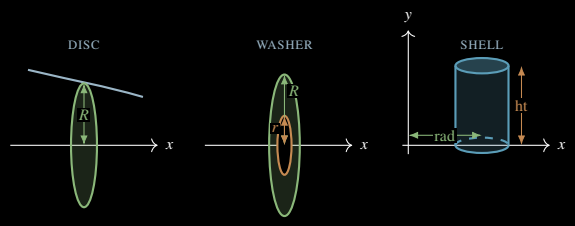
Discs and washers integrate in the axis variable; shells integrate in the other one. If solving the curve for the other variable is ugly, switch method rather than fight it.

THE THREE THINGS YOU SOLVE
1. **Intersections** — where the region starts and stops.
2. **The radius.** About $y=k$ the radius is $|k-y|$, not y. Draw the arrow from axis to curve.
3. **Solve for the other variable** when the slice direction needs it: $y=\sqrt{x} \Rightarrow x=y^2$.

Axis → variable → radius

Rotate about	Integrate	Radius
x-axis	dx	y-value of curve
y-axis	dy	x-value of curve
$y=k$	dx	$\lvert k-y\rvert$
$x=k$	dy	$\lvert k-x\rvert$

Shell (rotating about y-axis): radius $=x$, height $=f(x)$ (top curve − bottom curve). Shifted axis $x=k$: radius $=|k-x|$. Shell and disc always give the same total volume.



Worked (disc): $y=\sqrt{x}$ about the x-axis on $[0,4]$

1. TOUCHES THE AXIS ⇒ DISC
radius = the y-value: $R=\sqrt{x}$

↓

2. SQUARE & INTEGRATE
$V=\pi\int_0^4(\sqrt{x})^2dx=\pi\int_0^4 x\,dx=\pi\left[\frac{x^2}{2}\right]_0^4=8\pi$

Worked (washer): $y=x$ and $y=x^2$ about the x-axis

1. GAP TO THE AXIS ⇒ WASHER
on $[0,1]$ the line rides above the parabola: outer $R=x$, inner $r=x^2$

↓

2. SQUARE EACH SEPARATELY, THEN SUBTRACT
$[R]^2-[r]^2=x^2-x^4$ *(never square the difference: $(R-r)^2$ is a different — wrong — number)*

↓

3. INTEGRATE
$V=\pi\int_0^1(x^2-x^4)\,dx=\pi\left[\frac{x^3}{3}-\frac{x^5}{5}\right]_0^1=\pi\left(\frac{1}{3}-\frac{1}{5}\right)=\pi\cdot\frac{5-3}{15}=\frac{2\pi}{15}$

Worked (shell): same region about the y-axis

1. ABOUT y-AXIS, STAYING IN dx ⇒ SHELL
radius $=x$, height = top − bottom $=x-x^2$

↓

2. ASSEMBLE 2π (rad)(ht)
$V=2\pi\int_0^1 x(x-x^2)\,dx=2\pi\int_0^1(x^2-x^3)\,dx$

↓

3. EVALUATE
$2\pi\left[\frac{x^3}{3}-\frac{x^4}{4}\right]_0^1=2\pi\left(\frac{1}{3}-\frac{1}{4}\right)=2\pi\cdot\frac{1}{12}=\frac{\pi}{6}$

↓

4. CHECK BY THE OTHER METHOD
washers in dy — solve each curve for x: parabola $x=\sqrt{y}$ (farther right = outer), line $x=y$ (inner):

$$\pi\int_0^1\left[(\sqrt{y})^2-y^2\right]dy=\pi\left[\frac{y^2}{2}-\frac{y^3}{3}\right]_0^1=\pi\left(\frac{1}{2}-\frac{1}{3}\right)=\frac{\pi}{6}\ \checkmark$$

(shell and washer must agree — free error check)

***Worked (shifted axis):** $y=\sqrt{x}$ **on** $[0,4]$ **about** $y=-1$*

1. RADII FROM $|k-y|$
axis $y=-1$ sits *below* the region: outer $R=\sqrt{x}-(-1)=\sqrt{x}+1$, inner $r=0-(-1)=1$
(the gap from axis to region floor)

↓

2. WASHER SETUP — FOIL, THEN CANCEL
$(\sqrt{x}+1)^2-1=\left(x+2\sqrt{x}+1\right)-1=x+2\sqrt{x}$: $V=\pi\int_0^4\left(x+2\sqrt{x}\right)dx$

↓

3. EVALUATE
$\pi\left[\frac{x^2}{2}+\frac{4}{3}x^{3/2}\right]_0^4=\pi\left(8+\frac{32}{3}\right)=\frac{56\pi}{3}$

arc length

Length of a curve $y=f(x)$ on $[a,b]$ (Pythagoras on infinitely small pieces $\sqrt{dx^2+dy^2}$):

$$L=\int_a^b\sqrt{1+\left(\frac{dy}{dx}\right)^2}\,dx$$

Steps: find $f'(x)$ → square it → add 1 → square root → integrate. Doable problems are rigged so $1+(f')^2$ collapses to a perfect square.

***Worked:** $y=\frac{x^3}{6}+\frac{1}{2x}$ **on** $[1,2]$*

1. DIFFERENTIATE
$y'=\frac{x^2}{2}-\frac{1}{2x^2}$

↓

2. SQUARE & ADD 1 — SPOT THE FLIP
$1+(y')^2=1+\frac{x^4}{4}-\frac{1}{2}+\frac{1}{4x^4}=\frac{x^4}{4}+\frac{1}{2}+\frac{1}{4x^4}=\left(\frac{x^2}{2}+\frac{1}{2x^2}\right)^2$ *(adding 1 flips the middle sign: $(a-b)^2\to(a+b)^2$)*

↓

3. ROOT DROPS CLEANLY
$L=\int_1^2\left(\frac{x^2}{2}+\frac{1}{2x^2}\right)dx$

↓

4. EVALUATE
$\left[\frac{x^3}{6}-\frac{1}{2x}\right]_1^2=\frac{13}{12}-\left(-\frac{1}{3}\right)=\frac{17}{12}$

surface area of revolution

Spin the *curve* (not the region): each arc piece $ds=\sqrt{1+(y')^2}\,dx$ sweeps a hoop of circumference 2π(radius):

$$SA=\int_a^b 2\pi y\sqrt{1+(y')^2}\,dx\quad\text{(about the }x\text{-axis)}$$

About the y-axis the radius becomes x, so swap y for x and integrate in dy.

Worked:** $y=\sqrt{x}$ **on** $[0,4]$ **about the** x**-axis

1. THE PIECES
$y'=\frac{1}{2\sqrt{x}}$: $1+(y')^2=1+\frac{1}{4x}$

↓

2. COMBINE UNDER ONE ROOT
$\sqrt{x}\cdot\sqrt{1+\frac{1}{4x}}=\sqrt{x+\frac{1}{4}}$ *(the radius slides into the root — standard collapse)*

↓

3. INTEGRATE (linear inside)
$SA=2\pi\int_0^4\sqrt{x+\frac{1}{4}}\,dx=2\pi\cdot\frac{2}{3}\left(x+\frac{1}{4}\right)^{3/2}\Big|_0^4$

↓

4. EVALUATE
$\left(\frac{17}{4}\right)^{3/2}=\frac{17\sqrt{17}}{8}$, $\left(\frac{1}{4}\right)^{3/2}=\frac{1}{8}$: $\frac{4\pi}{3}\cdot\frac{17\sqrt{17}-1}{8}=\frac{\pi}{6}\left(17\sqrt{17}-1\right)\approx 36.2$

work (springs & pumping)

Work = force through a distance:

$$W=\int_a^b F(x)\,dx$$

- Constant force → simply $W=F\cdot d$.
- Varies with position → you must integrate.
- Units: N·m= J (metric), or ft·lb (US).

THE PROCESS — ALL THREE KINDS
1. Slice the object or the fluid.
2. Write the force on one slice.
3. Write the distance that same slice moves. Different slices move different distances.
4. Integrate force × distance.

Step 1 decoded — three kinds of work

A **spring** — force varies, distance is the same. Force $=kx$, so $W=\int_a^b kx\,dx$
B **lifting a cable or chain** — force varies as rope leaves the load. Slice the rope.
C **pumping fluid** — every slice has the same force law but a different distance to travel. Slice the fluid horizontally.

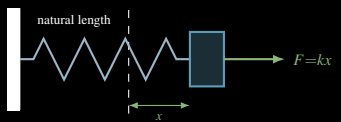
THE THREE THINGS YOU SOLVE
1. The constant k — from one given force and stretch: $F=kx$.
2. The weight of one slice — density × volume for fluid, mass per length for rope.
3. The distance that slice moves — measure from the slice to its destination, not to the origin.

Springs (Hooke's law)

Force to hold a spring stretched x past its natural length: $F(x)=kx$ (k= spring constant).
Work to stretch from a to b:

$$W=\int_a^b kx\,dx=\frac{k}{2}\left(b^2-a^2\right)$$

- **Find k first** from a known force at a known stretch: $k=F/x$.
- **Inverse problems** (force given work, or natural length given work): set the work integral equal to the given value, then solve.
- Measure x from the natural length, never from 0. A spring 0.1 m long stretched *to* 0.3 m has $x=0.2$.



Worked:** 40 **N holds a stretch of** 0.2 **m — work to stretch to** 0.3 **m

1. FIND k FROM THE GIVEN FORCE
$k=\frac{F}{x}=\frac{40}{0.2}=200$ N/m

↓

2. BOUNDS FROM NATURAL LENGTH
stretch runs $x=0$ to $x=0.3$

↓

3. INTEGRATE
$W=\int_0^{0.3}200x\,dx=100x^2\Big|_0^{0.3}=9$ J

Worked (inverse):** 2 **J stretches a spring** 0.1 **m — what force held it there?

1. SET THE WORK INTEGRAL EQUAL TO THE GIVEN
$\int_0^{0.1}kx\,dx=\frac{k}{2}(0.1)^2=0.005k=2$

↓

2. SOLVE FOR k
$k=400$ N/m

↓

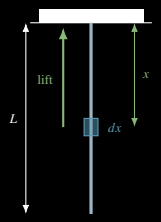
3. ANSWER THE ACTUAL QUESTION
$F = kx = 400(0.1) = 40$ N *(same machine, run backwards — work given, parameter wanted)*

Variable force / lifting

General rule: add up each slice's weight times the distance it rises.
For a hanging cable / rope of length L, weight density ρ (lb/ft): a slice of length dx at depth x weighs $\rho\,dx$ and rises a distance x, so

$$W = \int_0^L \rho\, x\, dx = \frac{\rho L^2}{2}$$

A constant load (bucket, weight) on the end adds (load)$\times L$ by **superposition**.



Worked: 50 ft cable, 2 lb/ft, wound to the top

1. SLICE
piece at depth x: weight $2\,dx$, rises x

↓

2. INTEGRATE
$$W = \int_0^{50} 2x\,dx = x^2\Big|_0^{50} = 2500 \text{ ft·lb}$$

↓

3. SUPERPOSE THE END LOAD
100 lb bucket on the end: $+100(50) = 5000$ $\Rightarrow$ total 7500 ft·lb

Partial pull — wind up only the top 20 ft: slices above 20 rise their own depth, slices below all rise exactly 20:
$W = \int_0^{20} 2x\,dx + \int_{20}^{50} 2(20)\,dx = 400 + 1200 = 1600$ ft·lb — the deep slices' lift stops growing.
Leaky bucket — 5 lb bucket $+40$ lb water lifted 20 ft, leaking to empty at the top: weight at height x is $5 + 40\left(1 - \frac{x}{20}\right) = 45 - 2x$:
$W = \int_0^{20} (45 - 2x)\,dx = 900 - 400 = 500$ ft·lb — a variable weight, same machinery.

Pumping fluid out of a tank

Slice the fluid into horizontal layers at height y, thickness dy. Each slice has volume $A(y)\,dy$ and weight $\gamma A(y)\,dy$, and must rise a distance $D(y)$ (the lift distance) to reach the spout:

$$W = \int \gamma A(y)\, D(y)\, dy$$

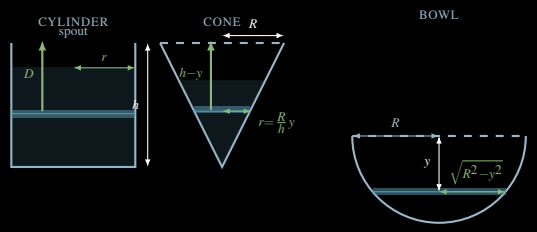
Weight density γ of water: 9800 N/m^3 (metric), or 62.4 lb/ft^3 (US).
3 steps:

1. Cross-sectional area $A(y)$ of the slice (use similar triangles or circle geometry).
2. Lift distance $D(y)$ up to the rim / spout.
3. Multiply and integrate over the fluid depth only.

Shape	Slice $A(y)$	from
Cylinder (rad r)	πr^2	constant
Inverted cone (h, top R)	$\pi\left(\frac{R}{h}y\right)^2$	similar △
Hemisphere bowl (rad R)	$\pi(R^2 - y^2)$	circle
Triangular trough (len L)	$Lw(y)$, w linear	similar △
Trapezoid trough (len L)	$L(w_0 + my)$	linear sides
Semicircular trough (len L)	$L \cdot 2\sqrt{R^2 - y^2}$	circle

Two traps:

- Integrate only over the actual *fluid* — not the empty top of a partly-full tank.
- If the spout sits *above* the rim, $D(y)$ includes that extra height.



Worked: full cone (apex down, $h = 10$ ft, top radius 4 ft) over the rim

1. SLICE AREA — SIMILAR TRIANGLES
radius grows with height: $r = \frac{4}{10}y = 0.4y$ $\Rightarrow$ $A(y) = \pi(0.4y)^2 = 0.16\pi y^2$ *(square the whole $0.4y$ — the classic slip)*

↓

2. LIFT DISTANCE
slice at height y rises to the rim: $D(y) = 10 - y$

↓

3. ASSEMBLE & INTEGRATE OVER THE FLUID
$$W = \int_0^{10} 62.4\,(0.16\pi y^2)(10 - y)\,dy = 9.984\pi \int_0^{10} (10y^2 - y^3)\,dy$$

↓

4. EVALUATE
$$9.984\pi\left[\frac{10y^3}{3} - \frac{y^4}{4}\right]_0^{10} = 9.984\pi\left(\frac{10000}{3} - 2500\right)$$
$$= 9.984\pi \cdot \frac{10000 - 7500}{3} = 9.984\pi \cdot \frac{2500}{3} = 8320\pi \approx 26{,}100 \text{ ft·lb}$$

The two traps, worked

Partly full — same cone, water only to depth 8: only the bounds change (the fluid is $y \in [0, 8]$; lift is still to the rim):

$$W = 9.984\pi \int_0^8 (10y^2 - y^3)\,dy = 9.984\pi\left[\frac{10y^3}{3} - \frac{y^4}{4}\right]_0^8$$
$$= 9.984\pi\left(\frac{5120}{3} - 1024\right) = 9.984\pi \cdot \frac{2048}{3} \approx 21{,}400 \text{ ft·lb}$$

Spout above the rim — full cone, spout 2 ft up: only D changes: $D = 12 - y$:

$$W = 9.984\pi \int_0^{10} (12y^2 - y^3)\,dy = 9.984\pi\left[4y^3 - \frac{y^4}{4}\right]_0^{10}$$
$$= 9.984\pi(4000 - 2500) = 14{,}976\pi \approx 47{,}000 \text{ ft·lb}$$

Cylinder (constant A) — radius 2 m, height 3 m, full, metric: $A = 4\pi$ always:

$$W = 9800 \cdot 4\pi \int_0^3 (3 - y)\,dy = 39{,}200\pi\left[3y - \frac{y^2}{2}\right]_0^3$$
$$= 39{,}200\pi\left(9 - \frac{9}{2}\right) = 39{,}200\pi \cdot \frac{9}{2} = 176{,}400\pi \approx 5.5 \times 10^5 \text{ J}$$

hydrostatic force

Pressure grows with depth: $P = \gamma d$ (depth d, weight density γ).
Horizontal plate — constant depth, so no integral:

$$F = P \cdot A = \gamma d A$$

Vertical plate — depth varies down the plate, so slice it into horizontal strips. A strip at depth $d(y)$ has width $w(y)$ and height dy, so the force on it is $dF = \gamma d(y)\, w(y)\, dy$. Add them up:

$$F = \gamma \int d(y)\, w(y)\, dy$$

THE PROCESS
1. Draw the plate and choose a depth variable.
2. Take a horizontal strip at depth h.
3. Write the strip width as a function of that depth.
4. Integrate $\rho g \times$ depth $\times$ width.

The only hard part is step 3: the width comes from the plate's geometry — constant for a rectangle, a similar-triangle ratio for a triangle, and $2\sqrt{r^2 - y^2}$ for a circle. Depth is measured downward from the surface, so check which way your variable runs before you integrate.

Vertical plate — strip method

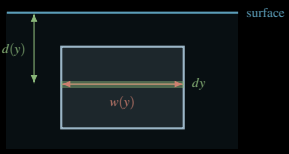
3 steps: pick a variable, then

1. strip width $w(y)$ from geometry,
2. strip depth $d(y)$ below the fluid surface,
3. integrate $\gamma \int d(y)\, w(y)\, dy$ over the plate.

Shape	Strip width $w(y)$
Rectangle (width w)	w (constant)
Isosceles triangle	linear in y (similar $\triangle$)
Semicircular window (rad R)	$2\sqrt{R^2-y^2}$

Depth is measured down from the fluid surface — not always the plate's top edge:

- Fully submerged window sits below the surface, so $d(y)$ carries a constant offset.
- Surface at the window's top: $d(y)$ is just the strip's position.



Worked: rectangular gate, 3 m wide × 2 m tall, top edge 1 m down

1. STRIP WIDTH
rectangle: $w(y)=3$, constant

↓

2. DEPTH & BOUNDS — FROM THE SURFACE
$y=$ depth below the surface: the plate occupies $y=1$ to 3 *(not 0 to 2 — the 1 m offset lives in the bounds)*

↓

3. INTEGRATE $\gamma\int d\,w\,dy$
$$F=9800\int_1^3 3y\,dy=29{,}400\cdot\frac{y^2}{2}\Big|_1^3=29{,}400\left(\frac{9}{2}-\frac{1}{2}\right)=29{,}400(4)=117.6\text{ kN}$$

Worked (triangle): vertex-down plate, base 4 m at the surface, height 3 m

1. WIDTH BY SIMILAR TRIANGLES
at depth y: $w(y)=4\left(1-\frac{y}{3}\right)$ *(full width at the surface, zero at the vertex)*

↓

2. ASSEMBLE
$$F=9800\int_0^3 y\cdot 4\left(1-\frac{y}{3}\right)dy=9800\int_0^3\left(4y-\frac{4y^2}{3}\right)dy$$

↓

3. EVALUATE
$$9800\left[2y^2-\frac{4y^3}{9}\right]_0^3=9800(18-12)=58.8\text{ kN}$$

Worked (semicircle): radius 2 m, flat edge at the surface

1. WIDTH FROM THE CIRCLE
$w(y)=2\sqrt{4-y^2}$ at depth y

↓

2. ASSEMBLE — SPOT THE U-SUB
$$F=9800\int_0^2 y\cdot 2\sqrt{4-y^2}\,dy$$ *($u=4-y^2$: the $y\,dy$ is sitting right there)*

↓

3. RUN THE U-SUB
$u=4-y^2$, $du=-2y\,dy \Rightarrow y\,dy=-\frac{1}{2}du$:
$$\int y\sqrt{4-y^2}\,dy=-\frac{1}{2}\int u^{1/2}du=-\frac{1}{2}\cdot\frac{2}{3}u^{3/2}=-\frac{1}{3}(4-y^2)^{3/2}$$

↓

4. EVALUATE — THE LOWER LIMIT CARRIES IT
$$19{,}600\left[-\tfrac{1}{3}(4-y^2)^{3/2}\right]_0^2=19{,}600\left(0-\left(-\tfrac{1}{3}\cdot 4^{3/2}\right)\right)=19{,}600\cdot\tfrac{8}{3}=\approx 52.3\text{ kN}$$ *($y=2$ gives 0 — the whole value rides on $y=0$)*

Horizontal plate — no integral: circular drain plate, radius 1 m, flat on the floor of a 4 m deep pool: constant depth, so $F=\gamma dA=9800\cdot 4\cdot\pi(1)^2=\approx 123$ kN.

improper integrals

Integrals with an infinite bound or an infinite discontinuity. Rewrite as a limit, integrate, then take the limit.

THE PROCESS
1. Find the bad point — an infinite bound, or where the integrand blows up.
2. Replace it with a limit variable.
3. Integrate normally.
4. Take the limit. If it is finite, the integral converges. If not, it diverges.

Step 1 decoded — three kinds

A **Type 1** — an infinite bound: $\int_1^\infty$

B **Type 2, endpoint** — the integrand blows up at a or b: $\int_0^1 \frac{dx}{\sqrt{x}}$

C **Type 2, interior** — the blow-up sits inside the interval. Split there first, and both halves must converge.

C is the trap: $\int_{-1}^1 \frac{dx}{x^2}$ looks routine and is divergent. Check the integrand for a zero denominator before you integrate anything.

Type 1 — infinite bound

$$\int_a^\infty f\,dx=\lim_{t\to\infty}\int_a^t f\,dx \qquad \int_{-\infty}^\infty f\,dx=\int_{-\infty}^c+\int_c^\infty$$

A two-sided ∞ must be split. It converges only if *both* halves do.

Worked: the canonical pair — $\int_1^\infty \frac{dx}{x^2}$ vs $\int_1^\infty \frac{dx}{x}$

1. REWRITE AS LIMITS
$$\lim_{t\to\infty}\int_1^t x^{-2}dx \quad \text{vs} \quad \lim_{t\to\infty}\int_1^t \frac{dx}{x}$$

↓

2. INTEGRATE
$$\left[-\tfrac{1}{x}\right]_1^t=1-\tfrac{1}{t} \quad \text{vs} \quad \left[\ln x\right]_1^t=\ln t$$

↓

3. TAKE THE LIMITS
$1-\frac{1}{t}\to 1$ converges; $\ln t\to\infty$ diverges

Same tails, opposite fates — $\frac{1}{x^2}$ dies fast enough, $\frac{1}{x}$ doesn't. That knife-edge is exactly the p-test below.

Type 2 — discontinuity in $[a,b]$

If f blows up at an endpoint (say b): $\int_a^b f\,dx=\lim_{t\to b^-}\int_a^t f\,dx$. If the blow-up is *interior* at c, split at c first.

Worked: endpoint blow-up — $\int_0^1 \frac{dx}{\sqrt{x}}$

1. LIMIT AT THE BAD ENDPOINT
$$\lim_{t\to 0^+}\int_t^1 x^{-1/2}dx$$

↓

2. INTEGRATE & EVALUATE
$$\left[2\sqrt{x}\right]_t^1=2-2\sqrt{t}$$

↓

3. LIMIT
$t\to 0^+$: 2 — converges *(infinite height, finite area)*

Worked: the interior trap — $\int_{-1}^1 \frac{dx}{x^2}$

1. THE NAIVE (WRONG) ROUTE
$\left[-\frac{1}{x}\right]_{-1}^1=-1-1=-2$ — negative area from a positive function: impossible

↓

2. SPOT THE INTERIOR BLOW-UP
$f\to\infty$ at $x=0$, *inside* the interval $\Rightarrow$ must split: $\int_{-1}^0+\int_0^1$

↓

3. TEST ONE HALF
$$\lim_{t\to 0^+}\int_t^1 x^{-2}dx=\lim\left(\tfrac{1}{t}-1\right)=\infty \Rightarrow \text{diverges}$$ *(one bad half sinks the whole integral)*

Converge or diverge?

- Limit is a finite number ⇒ converges (to that number).
- Limit is $\pm\infty$ or doesn't exist ⇒ diverges.
- ***p*-test**: $\int_1^\infty \frac{1}{x^p}\,dx$ converges iff $p>1$; $\int_0^1 \frac{1}{x^p}\,dx$ converges iff $p<1$.
- **Comparison**: $0 \le f \le g$. If $\int g$ converges so does $\int f$; if $\int f$ diverges so does $\int g$.

The two p-test flavors flip — big powers win at ∞, small powers win near 0. The worked pair above is the p-test at $p=2$ (converges) and $p=1$ (diverges); the $\sqrt{x}$ example is the 0-to-1 flavor at $p=\frac{1}{2}<1$. **Comparison example**: $\int_1^\infty e^{-x^2}\,dx$ — no elementary antiderivative, but $e^{-x^2} \le e^{-x}$ for $x \ge 1$ and $\int_1^\infty e^{-x}dx = e^{-1}$ converges ⇒ converges by comparison.

integration pitfalls

- The minus in [top−(bottom)] distributes to all terms.
- Power rule on a negative exponent moves it *toward* zero: $-3+1=-2$ (not -4).
- $\int x^{-1}\,dx = \ln|x|$ — not the power rule (would divide by 0).
- $(ab)^n = a^n b^n$ — *both* factors get the power.
- Always $F(b)-F(a)$, upper minus lower — never reversed.

Legend

Color & notes
functions $f, g,$ sin
variables x, y, n
numbers 0–9
operators $+ - =$
delimiters $( )$ $[ ]$
titles in white
labels · results · pitfalls
$\boxed{u}$ / $\boxed{du}$ substitution

Greek
constants (fixed values):
π ≈ 3.14159
γ fluid wt density
ρ weight per length
variable & operators:
θ angle (varies)
Δ "change in"
Σ sum of terms
notation, not values:
$\Theta\,\Omega\,\omega$ growth rate, like O

Calculus & relations
$\int$ integral
$\sum$ sum
lim limit
∞ infinity
$\to$ approaches
$\Rightarrow$ implies
$\Leftrightarrow$ iff
$\sqrt{\ }$ root
$\approx$ approx
$\not= \le \ge$ compare
$\pm \mp$ plus / minus
$\cdot \times \div$ mult / div
$\circ$ compose

Vectors
$\langle\,\rangle$ components
$\|\vec{v}\|$ magnitude
$\vec{v}$, $\hat{v}$ vector, unit
$\mathbf{i}\,\mathbf{j}\,\mathbf{k}$ basis
$\vec{u}\cdot\vec{v}$ dot product
proj projection

Logic & sets
$\forall$ for all
$\exists$ exists
$\neg$ not
$\wedge \vee$ and / or
$\oplus$ xor / sym. diff
$\in \notin$ in / not in
$\subseteq \subset$ subset
$\cup \cap$ union / inter.
$\setminus$ set minus
$\equiv$ equiv / congruent
$\mid$ divides / $|x|$ / s.t.
$\lceil\,\rceil$ ceiling
$\binom{n}{k}$ choose
deg degree
mod modulo